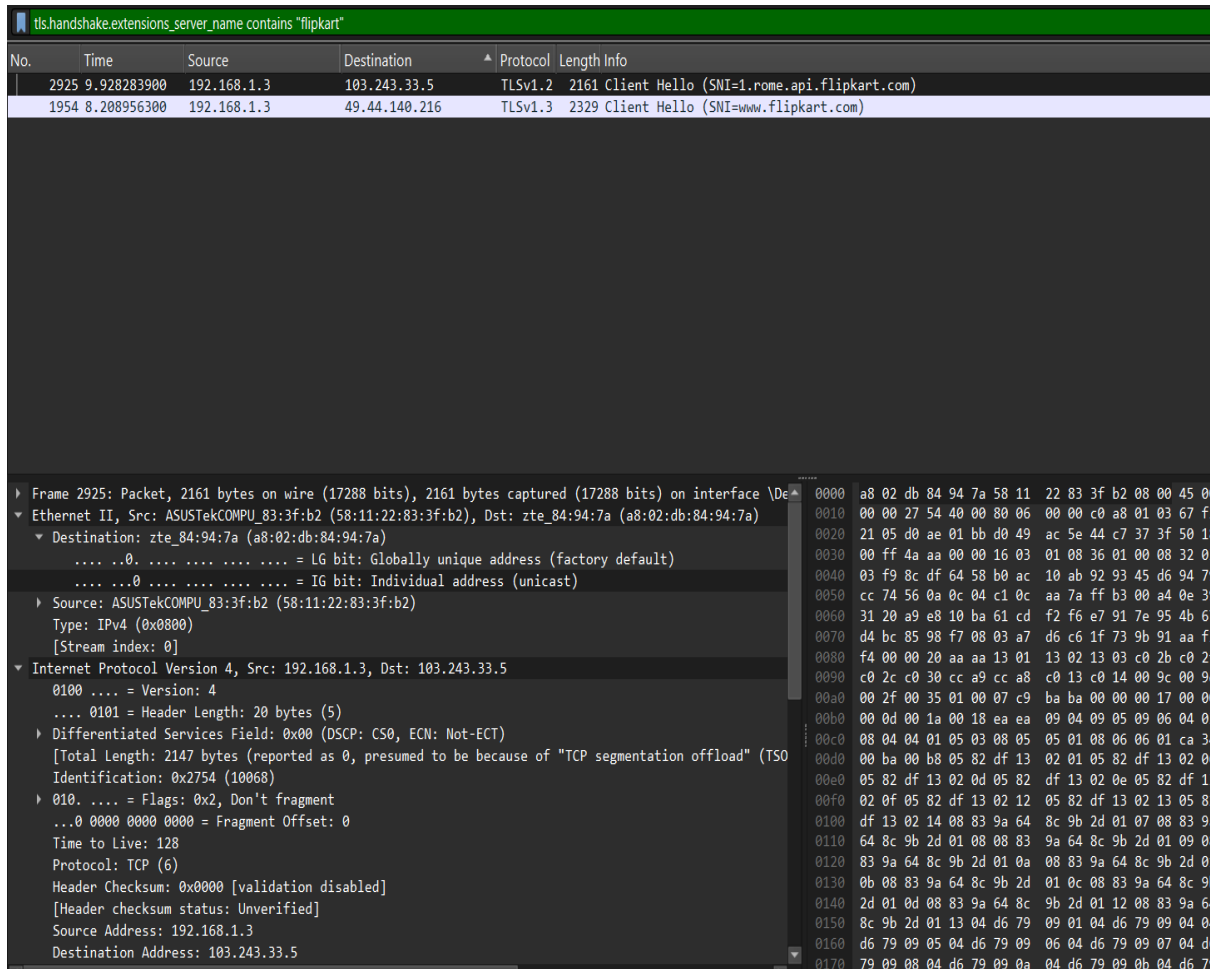


Day 2 Tcp/Ip Suite

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

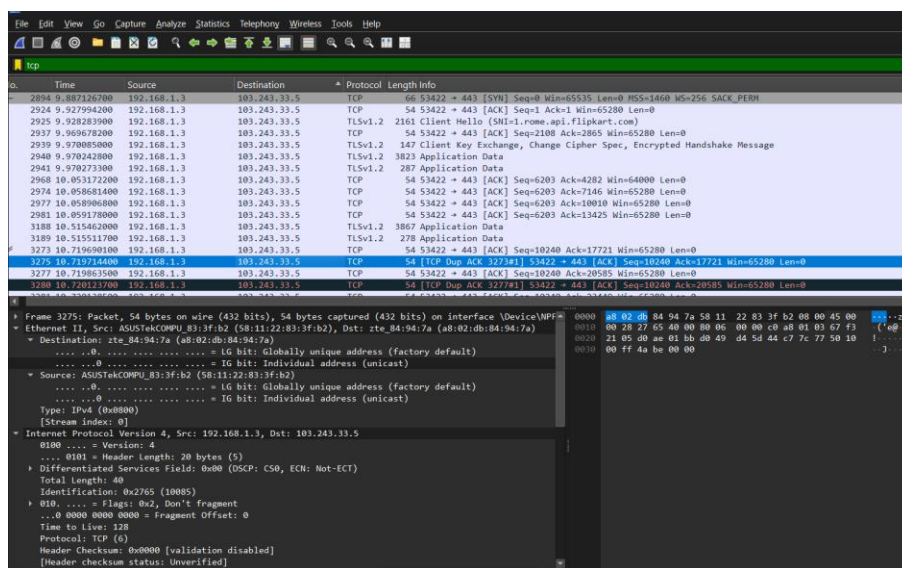


OSI/Model Layer	Wireshark Protocol
Application	TLSv1.2
Transport	TCP
Internet	IPv4
Network Access	Ethernet II

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

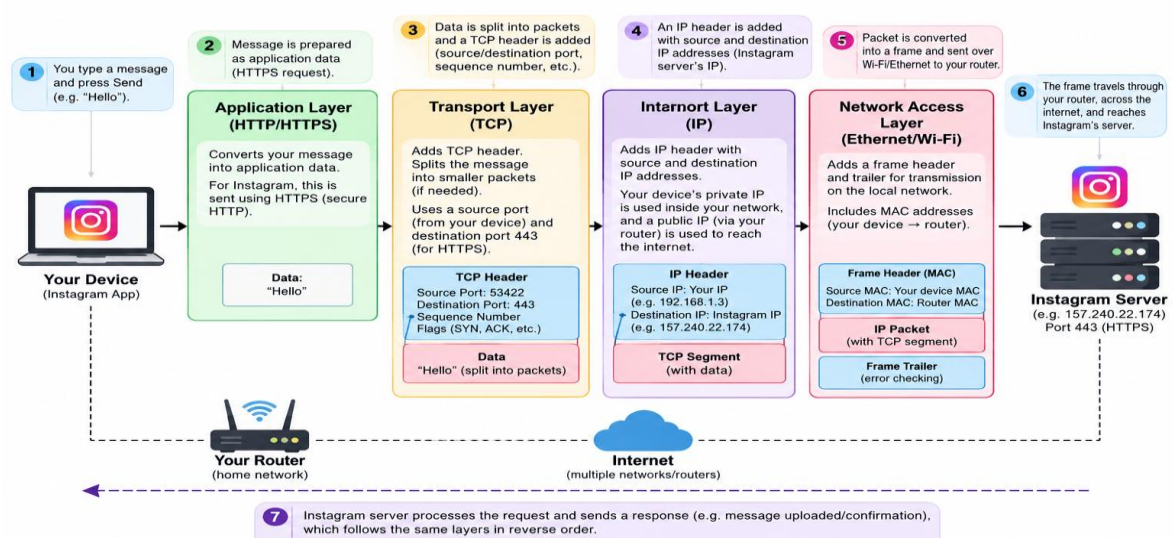
OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	Sending a WhatsApp message or placing a Zomato order
6. Presentation	Application	WhatsApp encrypting a message or Instagram compressing an image
5. Session	Application	Maintaining a WhatsApp login/session while using the app
4. Transport	Transport	TCP/UDP/QUIC carrying data for WhatsApp, YouTube, or Instagram
3. Network	Internet	IP routing when accessing WhatsApp or Zomato servers
2. Data Link	Network Access	Wi-Fi/Ethernet frames transferring data between your phone/laptop and router
1. Physical	Network Access	Wi-Fi radio signals or Ethernet electrical signals carrying the data

3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

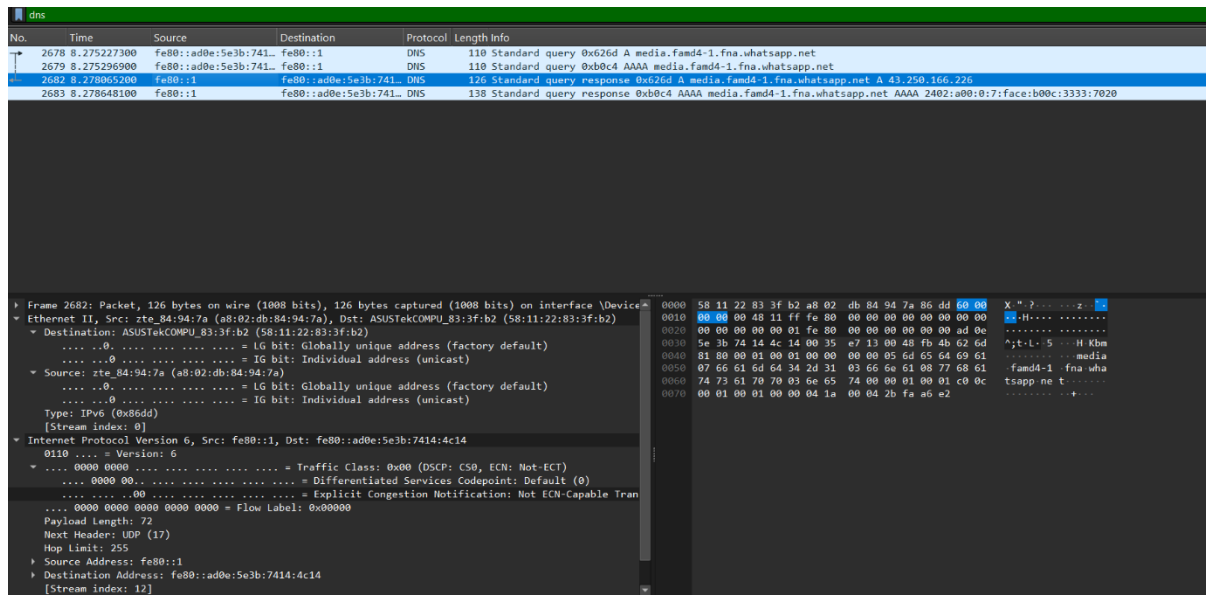


TCP Field	Explanation
Source Port	Identifies the application sending the TCP data.
Destination Port	Identifies the application receiving the data. Here, port 443 is used for HTTPS.
Sequence Number	Keeps track of the position of data in the TCP connection.
Acknowledgment Number	Indicates the next byte of data that the receiver expects.
Header Length	Indicates the size of the TCP header.
Flags	Control the TCP connection. For example, SYN starts a connection and ACK acknowledges data.
Window Size	Indicates how much data can be received before an acknowledgment is required.
Checksum	Detects errors in the TCP header and data.
Urgent Pointer	Points to urgent data when the URG flag is enabled.
Options	Provides additional TCP features such as MSS and window scaling.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.



TCP/IP Layer: Application Layer

How I identified it: I identified the packet as an Application-layer packet because Wireshark shows the protocol as **DNS**. The packet contains a WhatsApp domain (whatsapp.net).

Real-world action: This packet represents WhatsApp performing DNS resolution to find the server address needed for communication when sending a message.